

A Framework for Estimating Road-Blockage Probability from Conflict Events

A weakly-supervised, parametric impact model

Conflict-event → Road-Blockage (C2RB) model

ACLED, Afghanistan, 2017–2026 (69,655 geocoded events)

1 Problem and approach

Goal. Given a conflict event (or a set of events over a time window), estimate which nearby roads are *affected* and the *probability that each road is blocked*. The central modelling question is: **how large is an event’s impact “buffer zone”, and how does it differ by sub_event_type?**

The core difficulty — no ground truth. The ACLED export has no column that says “a road was blocked”. A fully supervised classifier therefore cannot be trained directly. Two naive alternatives both fail:

- *Pure assumption* (pick buffer radii by hand): not defensible, not reproducible.
- *Pure ML* (train a closure classifier): impossible without labels.

Our approach — a weakly-supervised parametric probability model. We build an explicit, interpretable probability model whose parameters are (a) **seeded from published literature** and (b) **calibrated against weak labels mined from the ACLED notes free text**. This is defensible, reproducible, and mirrors established practice: ACLED’s own *Conflict Exposure* product uses fixed event-type buffers (1/2/5 km) with Voronoi de-duplication, and a published conflict-impact model assigns per-event-type radii (Battles 25 km, Explosions 10 km, Protests/Riots 5 km) with type-specific decay shapes.

Pipeline. event $\rightarrow R_{\text{eff}} \rightarrow K(d) \rightarrow P_{\text{block}}(d) \xrightarrow{\text{noisy-OR} + \text{time decay}} P(\text{road blocked})$

2 The model

For an event e with sub-event-type s , geo-precision code $g \in \{1, 2, 3\}$, fatalities n_f , and civilian-targeting flag $c \in \{0, 1\}$:

2.1 Effective buffer radius — “how big is the zone?”

$$R_{\text{eff}}(e) = \sqrt{R_{\text{phys}}(s)^2 + R_{\text{geo}}(g)^2} \cdot M_{\text{sev}}(e) \quad (1)$$

- $R_{\text{phys}}(s)$ — physical/operational impact radius of the event type (km), literature-seeded per `sub_event_type`. E.g. IED 1 km, armed clash 15 km, protest 4 km.
- $R_{\text{geo}}(g)$ — *locational uncertainty* of where the event actually happened. ACLED geo-precision 1 $\approx$ town (1 km), 2 $\approx$ near a town / part of a region (5 km), 3 $\approx$ wider region (25 km). A low-precision event must spread its blockage probability over a larger uncertain area.
- **Quadrature combination** $\sqrt{R_{\text{phys}}^2 + R_{\text{geo}}^2}$. Both terms behave like independent Gaussian length-scales (physical spread and locational error); variances of independent Gaussians add, so their standard deviations add in quadrature. This is more principled than naive addition (which double-counts) and never smaller than either term.

The severity multiplier $M_{\text{sev}}(e)$ stretches the radius for deadlier and civilian-directed events, with diminishing returns so a 50-fatality battle does not produce a $50\times$ radius:

$$S(n_f) = \alpha(1 + n_f) + (1 - \alpha)(1 + \ln(1 + n_f)) \quad (2)$$

$$M_{\text{sev}}(e) = \text{clip}\left(1 + \kappa\left(\frac{S(n_f)}{S(n_{\text{ref}})} - 1\right), 0.5, 3.0\right) \cdot (1 + \gamma_{\text{civ}} c) \quad (3)$$

with defaults $\alpha = 0.15$ (mostly log growth), $\kappa = 0.25$, $\gamma_{\text{civ}} = 0.30$. The `clip` bounds keep outliers from exploding the buffer.

2.2 Distance-decay blockage probability — “how does it fall with distance?”

$$P_{\text{block}}(d, e) = P_0(s) \cdot K_s(d; R_{\text{eff}}) \quad (4)$$

- $P_0(s)$ — *peak blockage propensity*: the probability that an event of type s blocks a road at the epicentre. **This is the parameter calibrated from notes** (Sec. 3).
- $K_s(d) \in [0, 1]$ — distance-decay kernel, its *shape* chosen by the event type’s physics (Table 1).

Kernel	Formula	Used for	Rationale
Gaussian	$\exp\left(-\frac{d^2}{2\sigma^2}\right), \sigma = R_{\text{eff}}/2$	IED, suicide, grenade, shelling, airstrike	sharp, blast-like, fast decay
Exponential	$\exp\left(-\frac{d}{\lambda}\right), \lambda = R_{\text{eff}}/3$	armed clash / battles	heavier tail — fighting spreads along terrain/roads
Uniform	$\mathbf{1}\{d \leq R_{\text{eff}}\}$	protests, riots, territorial control	roughly constant over an area, then cut off
Point	$\mathbf{1}\{d \leq \max(R_{\text{eff}}, 0.3)\}$	targeted violence, arrests	effect confined to the site

Table 1: Distance-decay kernels, chosen per event type.

2.3 Combining many events — noisy-OR with temporal decay

A road segment near several events is blocked if *any* of them blocks it. Treating events as independent blocking attempts gives the **noisy-OR**, with an **exponential temporal decay** $\varphi(\Delta t) = 0.5^{\Delta t/\tau}$ (half-life $\tau = 30$ days) so older events fade:

$$P_{\text{segment}} = 1 - \prod_{e \in W} \left[1 - P_{\text{block}}(d_e, e) \cdot \varphi(\Delta t_e)\right] \quad (5)$$

over a time window W . ACLED-style Voronoi de-duplication can be layered on to avoid double-counting tightly-clustered events (noted as an extension).

2.4 From points to roads

We use the WFP/AGCHO Afghanistan road network (`afg_trs_roads_1_arazi_2023x_fixed_segmented.shp`): **27,991 LineString segments, ≈ 1 km each** (median 1,068 m), EPSG:4326, each carrying road name, class, average slope and altitude. Each segment is represented by its midpoint; d_e is the segment-to-event distance. Every segment within $r_{\max} = 3 R_{\text{eff}}$ of an event receives that event’s $P_{\text{block}}(d_e, e)$, and segments are aggregated by noisy-OR. The result is a per-segment blockage probability that rolls up to **named roads**. Without a road layer the same scoring runs over a regular lat/lon grid, producing a continuous blockage-risk raster (the fallback) — the code path is identical, only the target points differ.

3 Calibration & validation from notes (weak supervision)

3.1 Building a weak “road-disruption” label

The export has no “road blocked” column, so we create one from the `notes` free text. Reading the notes reveals **three distinct mechanisms** by which an event disrupts a road; each is captured by its own regular expression and the label is their union, `road_disruption = block  $\vee$  hazard  $\vee$  denial`.

Mechanism	Signal in notes	Coverage	Typical type
Blockage	“blocked/closed the highway”, “cut off the road”, “under siege”	0.14%	protests, sieges
Hazard	“roadside bomb”, “IED/landmine on the road”, “bridge destroyed”	2.36%	IED / landmine
Denial	“convoy ambushed”, “set up checkpoint”, “seized control of the road”	1.89%	armed clashes
Union	any of the above	4.35%	3,032 events

Table 2: The three road-disruption mechanisms mined from `notes`.

Crucially the label is derived **only from text, never from `sub_event_type`** (only $\sim 18\%$ of IEDs are “roadside”), so the per-type rates below are genuine signal, not a relabelling of the type. This three-mechanism distinction is itself a finding: *road hazard* (a mined road) and *deliberate blockage* (protesters on a highway) are different phenomena that a single “road blocked?” keyword would have missed.

3.2 Peak blockage propensity $P_0(s)$

$P_0(s) = P(\text{road_disruption} \mid s)$ is estimated with **Beta–Binomial empirical-Bayes shrinkage** (prior strength 50, anchored at the 4.35% global rate) so small-sample types (Grenade $n=176$, Suicide $n=289$) get stable estimates instead of noisy zeros:

$$\hat{P}_0(s) = \frac{k_s + a_0}{n_s + a_0 + b_0}, \quad a_0 = \bar{p} m, \quad b_0 = (1 - \bar{p}) m, \quad m = 50, \quad (6)$$

where k_s positives out of n_s events of type s , and $\bar{p}$ is the global rate. Selected results with Wilson 95% intervals (full table in `params.yaml`):

Headline result. Blockage propensity spans $\sim 90\times$ across types: IEDs and landmines — deliberately placed *on roads* to deny movement — dominate, while air/drone strikes (which target buildings and people) almost never produce road-disruption language. This empirically confirms that the buffer/blockage model **must** vary by `sub_event_type`.

sub_event_type	n	P_0 (shrunk)	95% CI
Remote explosive/landmine/IED	8,545	0.181	[0.174, 0.190]
Suicide bomb	289	0.098	[0.077, 0.148]
Armed clash	39,974	0.031	[0.029, 0.032]
Peaceful protest	1,454	0.019	[0.012, 0.026]
Shelling/artillery/missile attack	2,540	0.010	[0.007, 0.014]
Air/drone strike	6,249	0.002	[0.001, 0.003]

Table 3: Calibrated blockage propensity — the headline result.

3.3 Validation — does the signal generalise?

A logistic regression $\text{road_disruption} \sim \text{sub_event_type} + \log(1 + \text{fatalities}) + \text{geo_precision} + \text{civilian_targeting}$, evaluated with 5-fold cross-validation:

- **AUC** = 0.79 — event attributes alone discriminate road-disrupting events well above chance.
- **Brier** = 0.039 — *better* than the no-skill baseline (0.042), i.e. genuinely informative.
- **Well-calibrated** — predicted $\approx$ observed across all deciles (top decile predicts 0.19, observes 0.17). Calibration requires the plain model; `class_weight="balanced"` improves ranking but inflates probabilities (use isotonic/Platt calibration if class-weighting).

The model is both discriminative and calibrated, so the $P_0(s)$ values are trustworthy as probabilities, not merely rankings.

3.4 Other sanity checks (in the notebook)

- **Roads-within-1 km benchmark** — **passed**. With the real road network loaded, **78% of events fall within 1 km of a road**, independently matching the published $\sim 70\%$ figure — strong external validation.
- **Sensitivity analysis** over R_{phys} and decay-kernel choice shows how the affected-road footprint responds to assumptions (what is assumed vs. learned).
- **Severity & geo-precision** move the weak label in sensible directions.

4 Statistical methods used

Distance-decay kernels (Gaussian / exponential / uniform); quadrature combination of independent uncertainty scales; bounded log-linear severity scaling; noisy-OR probabilistic aggregation with exponential temporal decay; weak supervision via regex labelling of free text; Beta-Binomial empirical-Bayes shrinkage with Wilson confidence intervals; logistic regression with cross-validated AUC / Brier / calibration-curve diagnostics; (optional) Voronoi tessellation for overlap de-duplication and KDE for hotspot context.

5 Limitations & honest caveats

1. **Weak labels are a proxy, not ground truth.** `road_disruption` measures *reported* road-relevant language in `notes`, not verified physical closures. Reporting bias propagates into P_0 . Replace with real closure data (OCHA/Logistics Cluster access reports) when available — the calibration code is unchanged.
2. R_{geo} **cannot be learned from this data.** ACLED snaps all events at a named location to one shared centroid, so intra-location coordinate spread is structurally ~ 0 km. We therefore fall back to ACLED’s *documented* precision semantics (1/5/25 km). With finer coordinates, the empirical-spread estimator in the notebook would populate R_{geo} .

3. R_{phys} **and kernel shapes are assumptions** (literature-seeded), unlike P_0 (data-calibrated). The sensitivity analysis quantifies how much they matter; tune in `params.yaml`.
4. **Independence in noisy-OR** slightly over-counts spatially correlated events; Voronoi de-duplication (ACLED’s approach) is the recommended refinement.
5. **Static / impact-only**. This scores the impact of given events; it does not forecast where future events will occur.

6 Files

File	Purpose
<code>c2rb.py</code>	Framework core: weak-label regexes, calibration, radius/kernel/noisy-OR math, scorer
<code>params.yaml</code>	All tunable parameters: R_{phys} , decay class, R_{geo} , calibrated P_0 , severity & time constants
<code>conflict_road_blockage.ipynb</code>	Annotated end-to-end walkthrough: load → label → calibrate → score → map → sensitivity
<code>build_notebook.py</code>	Regenerates the notebook from cell definitions (diffable source of truth)
<code>requirements.txt</code>	Pinned dependencies for reproducibility
<code>methodology.tex</code> / <code>.pdf</code>	This document

Reproduce. Create the venv and `pip install -r requirements.txt`, then run the notebook top-to-bottom. The road and admin layers are already wired in `params.yaml`; set `paths.roads: null` to fall back to grid-raster mode.

7 References

- ACLED, *Conflict Exposure methodology* (buffer zones 1/2/5 km, event-type adjustment, Voronoi de-duplication).
<https://acleddata.com/methodology/conflict-exposure-methodology>
- ACLED, *Codebook* (event types, sub-event types, geo-precision codes) — `AcledCodebook.md`.
- *Applying Computational Engineering Modelling to Analyse the Social Impact of Conflict and Violent Events* (per-type impact radii & decay shapes), PMC12562815.
<https://pmc.ncbi.nlm.nih.gov/articles/PMC12562815/>
- *Proximity to roads shapes patterns of violence in North and West Africa* (~70% of violent events within 1 km of a road).
<https://anl.geog.ufl.edu/roads-violence-nwafrica/>
- Road network: WFP / AGCHO-NSIA Afghanistan segmented roads (`afg_trs_roads_1_arazi_2023x`); admin boundaries: AGCHO-NSIA Afghanistan `adm1/adm2` (2017).